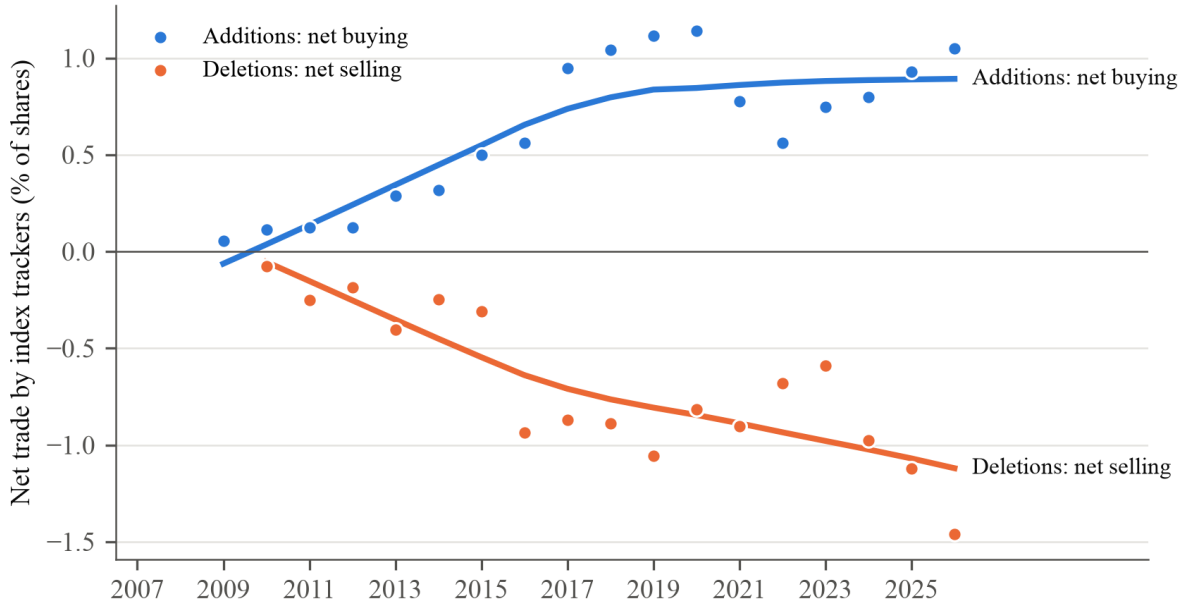
Figure 1. Net buying and selling by OMXSB index trackers, by year



Note. Net buying (additions) and net selling (deletions) by funds that track the OMXSB index, including tracker variants: the change in shares held between the last fund report before the announcement and the first report after the effective date, divided by shares outstanding (multiplied by 100). Fund holdings from LSEG Lipper; events with a share split between the two reports are excluded. Each point is the equal-weighted mean across events by year of the announcement; years without fund data are left out (312 events in total). The line is a LOWESS fit with bandwidth 0.8, as in Greenwood and Sammon (2025, Figure 1). Descriptive; no inference. Deletions caused by a takeover, merger, distressed share issue or a large change in shares outstanding (19 events) are excluded from the fund demand sample.

Table 1. Abnormal returns around OMXSB additions and deletions, by year, 2007–2026

	Additions				Deletions			
Year	**N**	**Ann.**	**Eff.**	**Total**	**N**	**Ann.**	**Eff.**	**Total**
2007	2	−0.53	5.99	7.39	6	0.28	−4.87	−7.22
2008	6	0.07	1.46	7.27	5	0.69	−3.04	−15.28
2009	9	2.41	3.57	13.87	10	3.37	3.07	−6.39
2010	5	−0.15	3.17	13.39	7	−0.58	−2.42	−6.23
2011	11	1.35	−2.98	2.82	10	−3.41	2.42	−2.72
2012	12	2.23	−0.84	−1.70	20	−0.94	1.01	3.98
2013	8	5.42	1.70	8.85	6	−2.93	−0.69	−7.78
2014	7	2.28	−6.18	−7.91	6	−3.17	−8.45	−14.82
2015	11	1.92	1.48	4.19	10	−2.20	0.20	−4.10
2016	16	−0.40	−0.98	−1.73	5	−0.43	−0.85	−3.59
2017	15	1.83	−1.58	6.15	9	−0.77	2.05	−0.32
2018	15	0.69	1.22	1.16	11	−3.47	−0.75	−1.17
2019	10	0.25	1.24	4.23	14	−0.07	0.92	1.56
2020	25	−3.31	2.11	2.48	13	−2.25	2.71	1.90
2021	17	1.67	−1.69	1.16	16	−2.95	−0.15	−4.16
2022	29	1.04	−0.64	0.74	21	−2.98	−0.76	−1.37
2023	13	−2.00	0.91	−0.87	17	1.17	−0.09	0.98
2024	17	1.74	−0.65	2.34	15	0.56	−0.55	−0.53
2025	17	1.88	−1.07	1.61	12	−0.30	−1.71	−1.54
2026	11	3.56	2.53	4.82	12	−1.02	−4.26	−6.55
Period means								
2007–11	33	0.96	1.16	8.52**	38	0.02	−0.17	−6.69***
2012–16	54	1.87**	−0.73	0.25	47	−1.69**	−0.79	−2.45
2017–21	82	−0.16	0.38	2.85***	63	−1.97**	0.86	−0.57
2022–26	87	1.20	−0.09	1.50*	77	−0.65	−1.26	−1.52
2007–16	87			3.39*	85			−4.34
2017–26	169			2.15***	140			−1.09

Note. Mean cumulative abnormal return in per cent (stock total return minus the OMX Stockholm Benchmark Gross Index, .OMXSBGI) by year of the effective date. Ann.: [AD−1, AD+1]; Eff.: [ED−1, ED+1]; Total: [AD−1, ED+1], where AD is the announcement date and ED the effective date. N counts events with a complete total window. Each year has at most two reviews, so yearly means carry no standard errors. Period rows: the same means pooled over five-year periods and decades, with stars from a restricted wild cluster bootstrap by review (Rademacher weights if $G \geq 12$, Webb six-point weights otherwise; 9,999 draws), where G is the number of reviews: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals in Table 2. 2007 contains only the July and December 2007 reviews, 2026 only the June review.

Table 2. Abnormal returns around OMXSB additions and deletions by period, 2007–2026

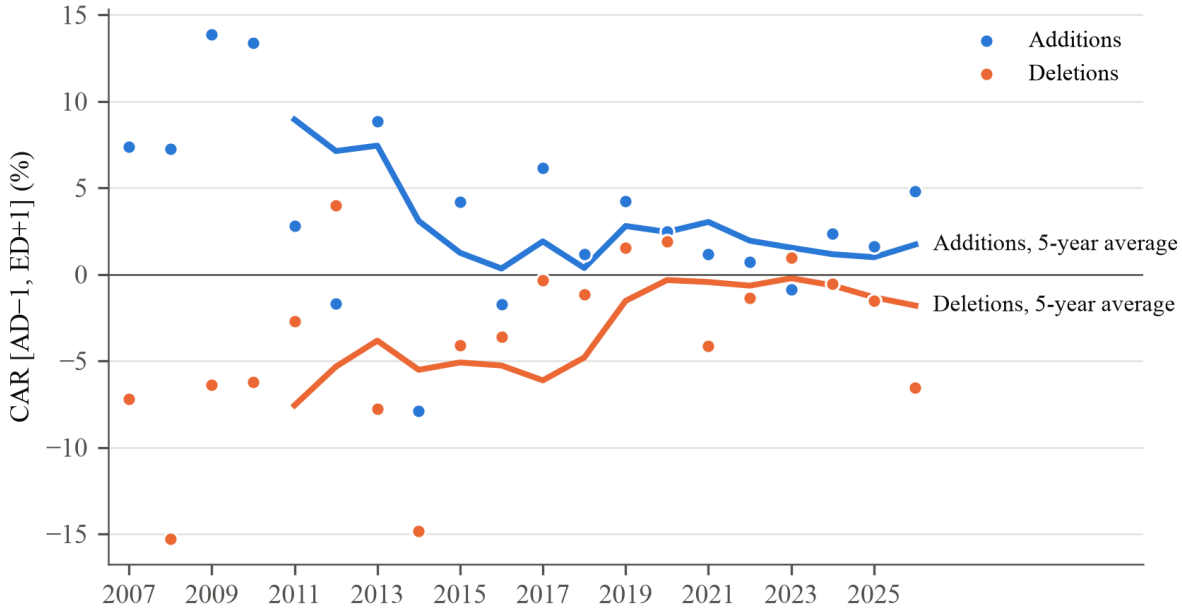
	2007–11	2012–16	2017–21	2022–26
Panel A. Additions				
Announcement [AD−1, AD+1]	0.96	1.87**	−0.16	1.20
	[−0.82, 2.98]	[0.07, 3.93]	[−3.05, 1.88]	[−0.39, 2.68]
Effective [ED−1, ED+1]	1.16	−0.73	0.38	−0.09
	[−1.74, 4.00]	[−2.96, 1.70]	[−1.45, 1.84]	[−1.17, 1.25]
Total [AD−1, ED+1]	8.52**	0.25	2.85***	1.50*
	[2.46, 14.74]	[−3.73, 5.30]	[1.26, 5.20]	[−0.10, 3.22]
Events	33	54	82	87
Reviews	9	10	10	9
Panel B. Deletions				
Announcement [AD−1, AD+1]	0.02	−1.69**	−1.97**	−0.65
	[−2.88, 3.30]	[−3.15, −0.86]	[−3.35, −0.55]	[−2.22, 0.96]
Effective [ED−1, ED+1]	−0.17	−0.79	0.86	−1.26
	[−4.09, 3.30]	[−5.81, 0.87]	[−0.31, 2.21]	[−2.88, 0.30]
Total [AD−1, ED+1]	−6.69***	−2.45	−0.57	−1.52
	[−11.45, −3.16]	[−13.53, 3.53]	[−3.56, 2.88]	[−4.22, 0.93]
Events	38	47	63	77
Reviews	9	10	10	9

Panel C. Total window [AD−1, ED+1], by decade

	2007–16	2017–26	Difference
Additions	3.39*	2.15***	−1.24
	[−0.25, 7.53]	[1.01, 3.39]	[−5.26, 2.54]
Events	87	169	
Deletions	−4.34	−1.09	3.25
	[−9.41, 0.80]	[−2.92, 0.82]	[−2.37, 8.56]
Events	85	140	

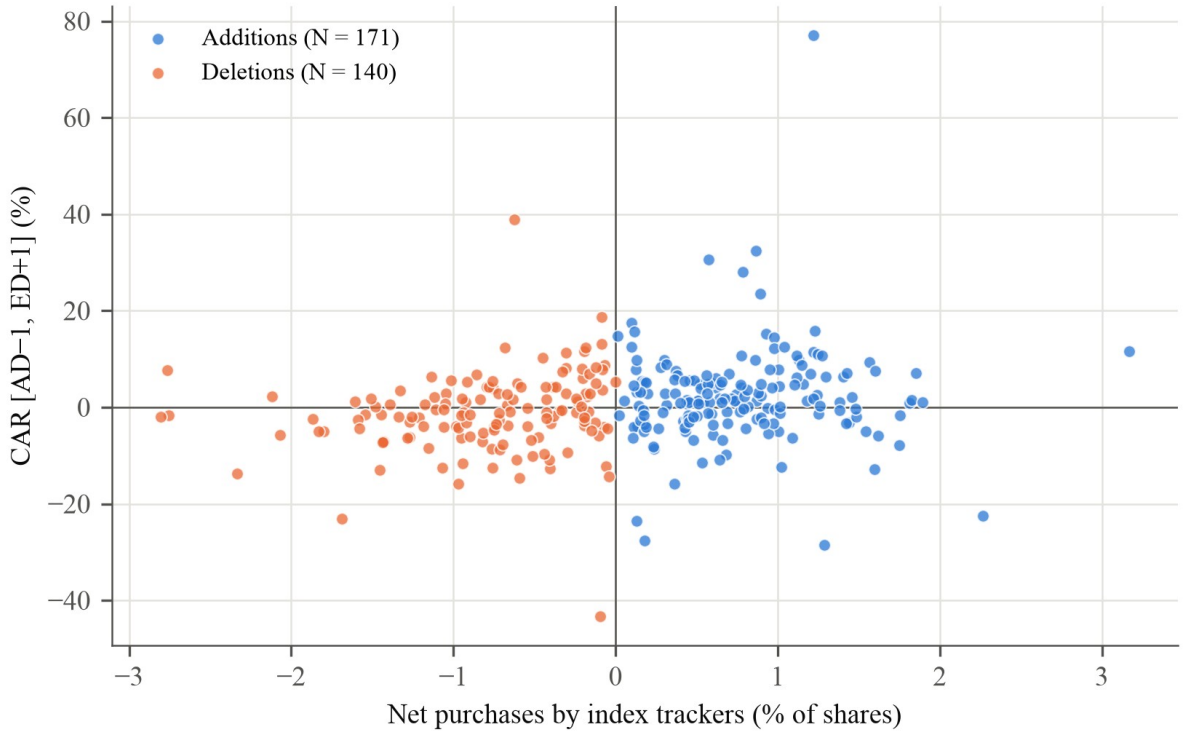
Note. Cumulative abnormal return in per cent: the stock's total return minus the return on the OMX Stockholm Benchmark Gross Index (.OMXSBGI), summed over the window. AD is the announcement date and ED the effective date. Each cell gives the mean across events. Inference: restricted wild cluster bootstrap by review (Rademacher weights if G ≥ 12, Webb six-point weights otherwise; 9,999 draws), where G is the number of reviews. Brackets: 95 per cent confidence intervals obtained by inverting the bootstrap test: they contain every value not rejected at the 5 per cent level; ∞ means that no value within 20 standard errors of the estimate is rejected. Stars come from the same bootstrap draws: * p < 0.10, ** p < 0.05, *** p < 0.01. Deletions keep their sign. Events are counted for the total window; the announcement and effective windows include up to three more events per period. Difference in Panel C: 2017–26 minus 2007–16, same inference. Sample: 481 changes in 39 semi-annual reviews with effective dates from 2 July 2007 to 1 June 2026 (Table A1).

Figure 2. Average index effect by year



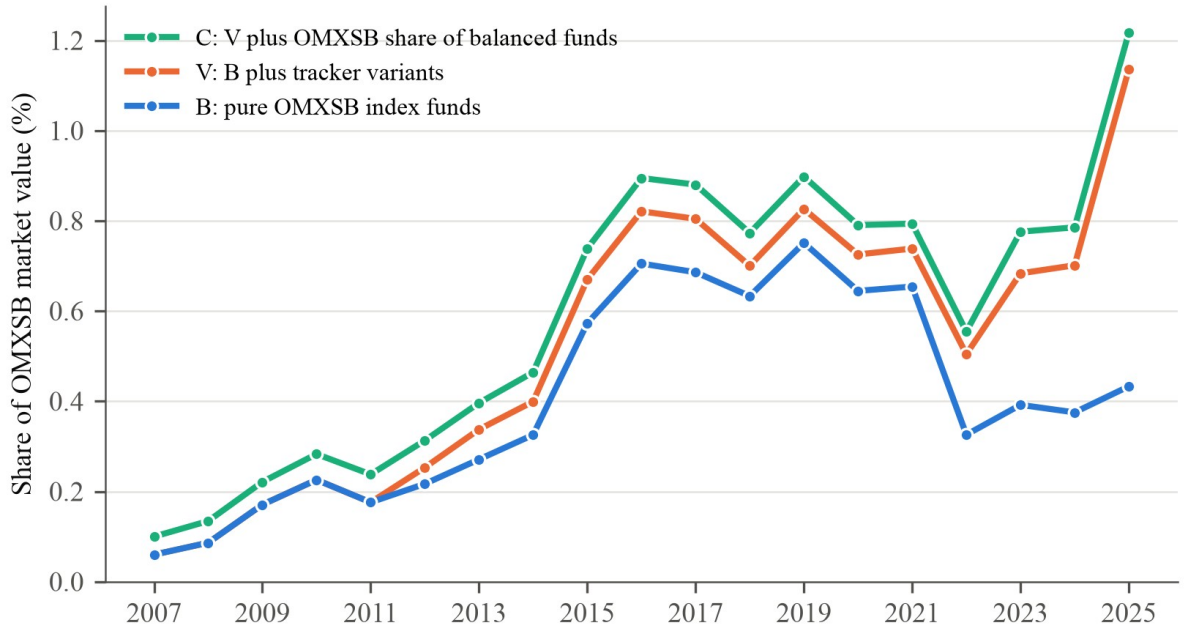
Note. Points: mean cumulative abnormal return from the day before the announcement to the day after the effective date ([AD−1, ED+1], in per cent, stock total return minus .OMXSBGI) by year of the effective date, as in Table 1. Lines: trailing 5-year moving averages of the yearly means (Greenwood and Sammon, 2025, Figure 2, use ten years over a 40-year sample). Descriptive; for inference see Table 2.

Figure 3. Net purchases by index trackers and the index effect



Note. Each point is one addition or deletion with fund holdings data and no share split between the fund reports (311 events, 2009–2026). Horizontal axis: net purchases by OMXSB index trackers including variants, in per cent of shares outstanding (negative for deletions). Vertical axis: cumulative abnormal return from the day before the announcement to the day after the effective date, in per cent. As in Greenwood and Sammon (2025, Figure 3). Descriptive; the multiplier estimates are in Table 3. Deletions caused by a takeover, merger, distressed share issue or a large change in shares outstanding (19 events) are excluded from the fund demand sample.

Figure 4. Share of the OMXSB index owned by index trackers, by year



Note. Assets of funds that track the OMXSB index at the end of each year (SHoF fund database), divided by the market value of the OMXSB index. The three lines are cumulative: B counts pure OMXSB index funds, V adds tracker variants, and C adds the OMXSB share of balanced funds, weighted by the OMXSB weight in their benchmark (Finansinspektionen). A fund counts only while its mandate documents show it tracking the index. Comparable to Greenwood and Sammon (2025, Figure 5). 2007–2025. Descriptive; no inference.

Table 3. Index effect decomposition: the multiplier M by period

	Additions				Deletions			
Period	**N**	**CAR**	**D̄**	**M**	**N**	**CAR**	**D̄**	**M**
Baseline window [AD−1, ED+1]								
2007–11	8	8.78	0.10	84.97 [13.05, 196.38]	7	0.19	−0.20	−0.93 [−58.86, 31.41]
2012–16	27	−1.94	0.42	−4.67 [−19.22, 2.20]	25	−2.07	−0.33	6.28 [−9.76, 37.14]
2017–21	58	3.29	1.00	3.29 [1.00, 6.87]	40	−0.20	−0.91	0.22 [−4.52, 3.49]
2022–26	78	2.28	0.78	2.94 [0.80, 5.37]	68	−1.66	−0.93	1.78 [0.00, 3.18]
2007–16	35	0.51	0.34	1.49 [−9.99, 17.50]	32	−1.58	−0.30	5.23 [−9.27, 28.01]
2017–26	136	2.71	0.87	3.11 [1.44, 5.14]	108	−1.12	−0.92	1.21 [−1.00, 2.80]
Extended window [AD−20, ED+1]								
2007–11	8	9.23	0.10	89.32 [60.57, 157.61]	7	−3.38	−0.20	16.75 [−46.06, 100.06]
2012–16	27	−0.94	0.42	−2.26 [−26.51, 14.25]	25	−4.04	−0.33	12.22 [−10.35, 55.79]
2017–21	58	4.69	1.00	4.69 [1.39, 9.11]	40	0.56	−0.91	−0.62 [−6.34, 6.16]
2022–26	75	6.62	0.79	8.34 [3.80, 13.27]	67	−2.91	−0.94	3.09 [−3.26, 7.30]
2007–16	35	1.39	0.34	4.03 [−14.10, 26.28]	32	−3.90	−0.30	12.88 [−7.41, 47.67]
2017–26	133	5.78	0.88	6.54 [3.51, 10.11]	107	−1.61	−0.93	1.74 [−2.96, 5.59]

Note. Following Greenwood and Sammon (2025, Table V), M is the mean cumulative abnormal return divided by the mean net demand of index trackers, CAR = M × D̄, estimated separately for additions and deletions and by period; the implied demand elasticity is ε = −1/M. CAR in per cent. D̄: mean net purchases (negative: net sales) by OMXSB index trackers including variants, in per cent of shares outstanding, between the fund report before the announcement and the report after the effective date. CAR keeps its sign (negative for deletions), so M is positive when prices move with index demand. Events with fund holdings data and no share split between the fund reports. Brackets: pairs cluster bootstrap by review (reviews drawn with replacement; 9,999 draws), percentile 95 per cent confidence intervals. No draw had a mean D of zero or below. Extended window: from 20 trading days before the announcement to the day after the effective date, as in their Table V. Deletions caused by a takeover, merger, distressed share issue or a large change in shares outstanding (19 events) are excluded from the fund demand sample.

Table 4. The multiplier M with and without controls

	Additions		Deletions	
	No controls	**Controls**	**No controls**	**Controls**
M, 2007–11	84.97 [11.98, 157.95]	79.68 [2.94, 156.41]	−0.93 [−16.84, 14.98]	2.17 [−12.48, 16.81]
M, 2012–16	−4.67 [−12.91, 3.57]	−1.96 [−9.55, 5.63]	6.28 [−15.51, 28.07]	4.49 [−16.87, 25.84]
M, 2017–21	3.29 [0.27, 6.31]	2.88 [−0.40, 6.16]	0.22 [−4.04, 4.47]	−0.02 [−3.96, 3.92]
M, 2022–26	2.94 [0.24, 5.65]	2.57 [0.92, 4.21]	1.78 [−0.46, 4.02]	1.93 [−0.47, 4.33]
Differences				
2007–11 − 2022–26	82.03 [−42.16, 140.25]	77.11 [−∞, 587.77]	−2.71 [−∞, ∞]	0.24 [−∞, ∞]
2012–16 − 2022–26	−7.61* [−18.81, 1.16]	−4.53 [−14.24, 4.34]	4.50 [−14.44, 53.57]	2.56 [−14.14, 48.72]
2017–21 − 2022–26	0.35 [−4.13, 5.08]	0.31 [−3.61, 5.02]	−1.57 [−7.37, 3.37]	−1.95 [−7.67, 2.96]
Events	171	163	140	131
Reviews	32	31	29	29
Controls	No	Yes	No	Yes

Note. Following Greenwood and Sammon (2025, Table VI), we estimate $CAR_i = \Sigma_k M_k \cdot 1\{\text{period } k\} \cdot \bar{D}_k + b'X_i + e_i$ separately for additions and deletions, where CAR_i is the cumulative abnormal return over [AD−1, ED+1] in the direction of index trading, $\bar{D}_k$ is the mean net demand of OMXSB index trackers in period k (per cent of shares outstanding), and X_i holds turnover, size, the Wurgler–Zhuravskaya arbitrage-risk measure and analyst coverage, measured before the announcement and centred. Without controls M_k equals the ratio in Table 3. Brackets under M: 95 per cent confidence intervals from standard errors clustered by review and t critical values with G − 1 degrees of freedom (no bootstrap). Differences: restricted wild cluster bootstrap by review (Rademacher weights if G ≥ 12, Webb six-point weights otherwise; 9,999 draws), where G is the number of reviews; brackets are 95 per cent confidence intervals obtained by inverting the bootstrap test: they contain every value not rejected at the 5 per cent level; ∞ means that no value within 20 standard errors of the estimate is rejected; stars from the same draws: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The 2007–11 period has few events with fund data, so its difference is imprecise. Deletions caused by a takeover, merger, distressed share issue or a large change in shares outstanding (19 events) are excluded from the fund demand sample.

Table 5. Net buying and selling by fund group around OMXSB index changes

	2007–11	2012–16	2017–21	2022–26	2017–26
Panel A. Additions: net buying					
OMXSB index trackers	0.10	0.42	1.00	0.78	0.87
	[0.06, 0.15]	[0.19, 0.64]	[0.88, 1.12]	[0.52, 1.03]	[0.72, 1.03]
Balanced funds, OMXSB share	0.00	0.03	0.09	0.07	0.08
	[0.00, 0.00]	[−0.01, 0.06]	[0.07, 0.11]	[0.06, 0.09]	[0.07, 0.09]
Other passive funds	−0.09	0.27	0.38	0.43	0.41
	[−0.57, 0.39]	[0.12, 0.41]	[0.17, 0.58]	[0.18, 0.67]	[0.26, 0.55]
Active funds	0.92	−0.03	−0.42	0.03	−0.16†
	[−0.78, 2.63]	[−0.92, 0.87]	[−0.86, 0.03]	[−0.39, 0.45]	[−0.47, 0.16]
All funds	0.94	0.68	1.04	1.30	1.19
	[−1.14, 3.01]	[−0.49, 1.86]	[0.57, 1.52]	[0.54, 2.07]	[0.76, 1.63]
Events	8	27	58	78	136
Reviews	5	8	10	9	19
Panel B. Deletions: net selling					
OMXSB index trackers	0.20	0.33	0.90	0.93	0.92
	[−0.27, 0.68]	[0.06, 0.60]	[0.80, 1.01]	[0.64, 1.23]	[0.75, 1.09]
Balanced funds, OMXSB share	0.00	0.01	0.07	0.10	0.09
	[−0.01, 0.00]	[−0.02, 0.04]	[0.06, 0.09]	[0.08, 0.13]	[0.07, 0.11]
Other passive funds	0.17	0.08	−0.16	−0.11	−0.13
	[−0.36, 0.71]	[−0.14, 0.29]	[−0.33, 0.00]	[−0.30, 0.07]	[−0.25, −0.02]
Active funds	−1.76	0.69	0.08	0.60	0.40†
	[−8.88, 5.36]	[−1.19, 2.57]	[−0.99, 1.15]	[0.13, 1.07]	[−0.09, 0.86]
All funds	−1.39	1.10	0.89	1.52	1.29
	[−7.66, 4.89]	[−0.83, 3.03]	[−0.35, 2.13]	[0.96, 2.09]	[0.73, 1.84]
Events	7	25	41	68	109
Reviews	3	7	10	9	19

Note. Mean change in shares held by each fund group between the fund report before the announcement and the report after the effective date, in per cent of shares outstanding, signed in the direction of index trading: positive numbers are purchases at additions and sales at deletions. Fund holdings from LSEG Lipper; events with a share split between the reports are excluded. OMXSB index trackers = pure index funds plus tracker variants. Balanced funds count with the OMXSB weight in their benchmark. Other passive funds are index funds that do not track the OMXSB, Swedish and foreign-domiciled. All funds is the sum of the groups. Brackets: 95 per cent confidence intervals from standard errors clustered by review and t critical values with G − 1 degrees of freedom (no bootstrap), except † where the interval is from a restricted wild cluster bootstrap by review (Rademacher weights if G ≥ 12, Webb six-point weights otherwise; 9,999 draws), where G is the number of reviews (95 per cent confidence intervals obtained by inverting the bootstrap test: they contain every value not rejected at the 5 per cent level; ∞ means that no value within 20 standard errors of the estimate is rejected). Comparable to Greenwood and Sammon (2025, Table VII), who compare index trackers with all 13F institutions. Deletions caused by a takeover, merger, distressed share issue or a large change in shares outstanding (19 events) are excluded from the fund demand sample.

Table 6. The index effect with and without overlapping MSCI Sweden changes

	All events		Excluding overlap		Overlap only	
Period	**CAR**	**N**	**CAR**	**N**	**CAR**	**N**
Panel A. Additions						
2007–11	8.52** [2.46, 14.71]	33	7.75** [2.57, 13.47]	31	20.51	2
2012–16	0.25 [−3.71, 5.37]	54	−0.27 [−3.53, 3.65]	50	6.79	4
2017–21	2.85*** [1.29, 5.19]	82	2.26*** [1.03, 3.89]	75	9.17	7
2022–26	1.50* [−0.14, 3.26]	87	0.73 [−0.77, 2.32]	82	14.16	5
2017–26	2.15*** [1.03, 3.39]	169	1.46*** [0.50, 2.47]	157	11.25	12
Panel B. Deletions						
2007–11	−6.69*** [−12.09, −3.16]	38	−6.69*** [−11.37, −3.18]	38		0
2012–16	−2.45 [−13.55, 3.51]	47	−1.49 [−10.65, 3.60]	45	−23.87	2
2017–21	−0.57 [−3.53, 2.93]	63	−0.07 [−2.64, 3.08]	60	−10.46	3
2022–26	−1.52 [−4.23, 0.92]	77	−1.44 [−4.28, 1.25]	73	−2.92	4
2017–26	−1.09 [−2.91, 0.82]	140	−0.82 [−2.65, 1.04]	133	−6.15	7

Panel C. Multiplier M, baseline window

	N	CAR (%)	D̄ (%)	M
Additions, 2017–21, all events	58	3.29	1.00	3.29 [0.98, 6.85]
Additions, 2017–21, excluding overlap	52	2.62	1.01	2.60 [0.60, 5.27]
Additions, 2022–26, all events	78	2.28	0.78	2.94 [0.80, 5.31]
Additions, 2022–26, excluding overlap	73	1.47	0.74	1.98 [0.64, 3.98]
Additions, 2017–26, all events	136	2.71	0.87	3.11 [1.48, 5.13]
Additions, 2017–26, excluding overlap	125	1.95	0.85	2.28 [0.98, 3.97]
Deletions, 2017–21, all events	40	−0.20	−0.91	0.22 [−4.56, 3.53]

Deletions, 2022–26, all events	68	−1.66	−0.93	1.78 [0.00, 3.19]
Deletions, 2017–26, all events	108	−1.12	−0.92	1.21 [−1.02, 2.80]
Deletions, 2017–26, excluding overlap	102	−0.71	−0.91	0.78 [−1.46, 2.40]

Note. An event overlaps when the same company is added to or deleted from the MSCI Sweden index within ten trading days of the OMXSB effective date. Panels A and B: mean cumulative abnormal return over [AD−1, ED+1] in per cent; All events equals Table 2. Inference: restricted wild cluster bootstrap by review (Rademacher weights if G ≥ 12, Webb six-point weights otherwise; 9,999 draws), where G is the number of reviews; brackets are 95 per cent confidence intervals obtained by inverting the bootstrap test: they contain every value not rejected at the 5 per cent level; ∞ means that no value within 20 standard errors of the estimate is rejected; * p < 0.10, ** p < 0.05, *** p < 0.01. Overlap only: descriptive mean, no inference. Panel C: M as in Table 3; brackets from a pairs cluster bootstrap by review (reviews drawn with replacement; 9,999 draws), percentile 95 per cent confidence intervals. Deletions in 2017–21 and 2022–26 are shown only for all events because they have three overlapping events each.

Table A2. Minimum detectable effects

Test	Estimate in	SE	Reviews	MDE
Fall in the addition effect, 2022–26 minus 2017–21	Table 2	1.062	38	3.06
Fall in the addition effect, 2017–26 minus 2007–16	Table 2	1.819	38	5.23
Addition effect, 2017–26	Table 2		19	1.62
Addition effect, 2022–26	Table 2		9	2.35
Deletion effect, 2017–26	Table 2		19	2.55
Change in M for additions, 2017–21 minus 2022–26	Table 4	1.988	32	5.75

Note. Minimum detectable effect for a two-sided 5 per cent test with 80 per cent power: MDE = (t_0.975 + t_0.80) × SE, with G − 1 degrees of freedom, where SE is the standard error clustered by review and G the number of reviews. Analytic, no bootstrap. Units: per cent abnormal return, except the change in M (per cent return per per cent of shares). SE is left blank where the MDE was computed directly from the estimate's file.

Table A3. The multiplier M with net demand from Finansinspektionen fund data, 2019–2026

				Finansinspektionen		LSEG Lipper	
Sample	**N**	**Reviews**	**CAR (%)**	**D̄ (%)**	**M**	**D̄ (%)**	**M**
Additions, excl. split events	106	15	2.43	1.14	2.12	0.88	2.75
					[0.87, 3.79]		[1.10, 5.04]
Additions, incl. split events	114	15	2.20	1.10	2.01		
					[0.76, 3.55]		
Deletions, excl. split events	89	15	−1.05	−1.16	0.91	−0.94	1.12
					[−1.21, 2.21]		[−1.51, 2.83]
Deletions, incl. split events	94	15	−1.33	−1.15	1.16		
					[−0.69, 2.38]		

Note. Not declared in advance; a check on the fund demand measure. Finansinspektionen: net purchases by funds classified as OMXSB index funds from their quarterly holdings in Finansinspektionen's fund data, counted only when classified so in both the quarter before and the quarter after the change, 2019–2026. LSEG Lipper: the measure used in Table 3, on the same events. CAR over [AD−1, ED+1], keeping its sign. M = mean CAR / mean D̄. Brackets: pairs cluster bootstrap by review (reviews drawn with replacement; 4,999 draws), percentile 95 per cent confidence intervals. The LSEG measure is not available for events with a share split between the fund reports.